

EXP 10: IMPLEMENT EXPECTATION AND MAXIMATION ALGORITHM

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AIM:

To implement the Expectation-Maximization (EM) algorithm in Python for clustering data using a Gaussian Mixture Model.

PROCEDURE

1. Import the required libraries and load the dataset into the program.
2. Create a Gaussian Mixture Model with the required number of clusters.
3. Train the model using the Expectation-Maximization algorithm.
4. Predict the cluster labels for the dataset.
5. Display the predicted labels and accuracy to evaluate the clustering performance.

PROGRAM:

```
# Expectation-Maximization (EM) Algorithm using Gaussian Mixture Model
```

```
from sklearn.datasets import load_iris
from sklearn.mixture import GaussianMixture
from sklearn.metrics import accuracy_score
from scipy.stats import mode
import numpy as np
```

```
# Load dataset
iris = load_iris()
X = iris.data
y = iris.target
```

```

# Create EM model

gmm = GaussianMixture(n_components=3, random_state=42)

# Train model

gmm.fit(X)

# Predict clusters

clusters = gmm.predict(X)

# Map clusters to actual class labels

labels = np.zeros_like(clusters)

for i in range(3):
    labels[clusters == i] = mode(y[clusters == i], keepdims=True).mode[0]

# Display results

print("Predicted Labels:")

print(labels)

print("\nAccuracy:")

print(accuracy_score(y, labels) * 100, "%")

```

OUTPUT:

```

Predicted Labels:
[0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 2 1 2 1 2 1
 1 1 1 2 1 1 1 1 1 2 1 1 1 1 1 1 1 1 1 1 1 1 1 1 2 2 2 2 2 2 2 2 2
 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2
 2 2]

Accuracy:
96.66666666666667 %

```

RESULT:

The Expectation-Maximization (EM) algorithm was successfully implemented in Python. The Gaussian Mixture Model clustered the dataset effectively, and the clustering performance was evaluated using the predicted labels and accuracy.